

Q1:JUN 2010 P2

(2)

(a) Define

1. *work*,

2. *power*.

[2]

- (b) By reference to equations of motion, derive an expression for the kinetic energy, E_k , of an object of mass, m , moving at speed, V , from rest.

Q2:JUN 2007 P3

2

(a) (i) Define work.

- (ii) A girl carrying a 10 kg bundle of firewood walks over a distance of 2.0 km.

Comment on the work done by the weight of the bundle.

[4]

- (b) A tennis ball dropped from rest onto a hard flat surface bounces to about **three-quarters** of its original height.

- (i) Neglecting air resistance, show that the velocity of the ball after falling through a height h is given by $v = \sqrt{2gh}$.

- (ii) Calculate the percentage of the mechanical energy lost in each bounce. Account for the lost energy.

- (iii) Practically, the ball bounces back to less than three-quarters of its initial height.

Suggest a reason for this.

[7]

Q3:JUN 2008 P3

- 2 (a) (i) Define work done.
- (ii) Explain why a stationary student holding a 20 kg bag of mealie-meal at a height of 1.50 m above the ground does not do mechanical work. [4]
- (b) A steady force is applied to a body of mass, m , with initial velocity, u , for a time, t , increasing its velocity to, v .
- Derive an expression for the
- (i) impulse given to the object,
- (ii) force applied to the object,
- (iii) average velocity of the object,
- (iv) distance moved while the force was applied,
- (v) work done by the force. [8]
- (c) Comment on the relationship between work done on an object and the change of energy of the object. [1]
- (d) A shop escalator at an angle of 45° to the horizontal has steps moving at 0.50 ms^{-1} between two floors of a building 8.0 m apart.
- (i) Given that the average mass of an adult human being is 70 kg, calculate the kinetic and gravitational potential energy the escalator has to give each of its passengers.
- (ii) The height of a step is 20 cm and two people can stand on each step.
- Calculate the maximum power required for the escalator motor given that a power of 1.2 kW is necessary to run the escalator with no load. [7]

Q4: NOV 2008 P3

- 1 (c) Fig. 1.1 shows a body of mass 50 kg moving down a rough incline with a constant velocity v .

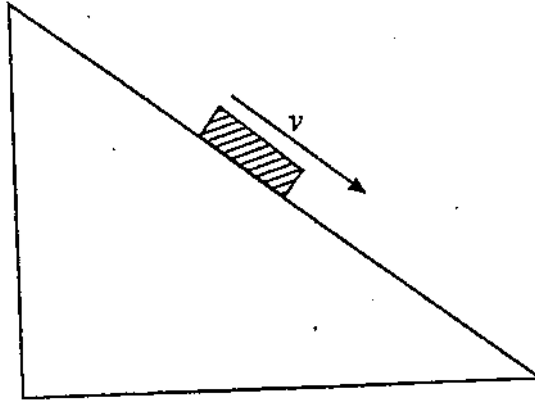


Fig. 1.1

- (i) Copy Fig. 1.1 and show all the forces acting on the body.
- (ii) Draw a triangle of forces for Fig. 1.1.

Q5: NOV 2009 P3

- 1 (a) (i) Define *work*.
- (ii) Derive from the definition in (i), the equation for the change in potential energy near the earth's surface.
- (iii) A cubic wooden block of mass 3.0 kg and side 0.75 m, lying horizontally on one of its faces is tilted through 30° as in Fig. 1.1.

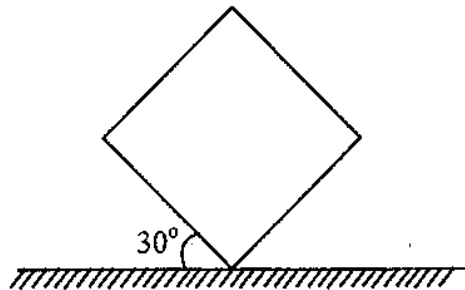


Fig. 1.1

Determine the change in potential energy.

[7]

Q6:NOV 2012 P3

- 4 (a) (i) Define *power*.
- (ii) Using the definition in (i) show that power, P , is given by $P = Fv$, where F is the force and v is the velocity.
- (b) A metal block of mass 5 kg is inclined at an angle of 30° to the horizontal. The block slides, from rest, down the plane a distance of 100 cm in 2.0 s.
- (i) Sketch a free body diagram for the block.
- (ii) Calculate the frictional force between the block and the plane when the block is sliding.
- (iii) Calculate the work done by the block.

[3]

[7]

Q7:JUN 2015 P3

- 2 (a) (i) State the relationship between *power* and *energy*.
- (ii) Deduce the base units of power.
- (b) At a school science exhibition, a student designed a model of a hydroelectric power station shown in Fig. 2.1 (not to scale).

[3]

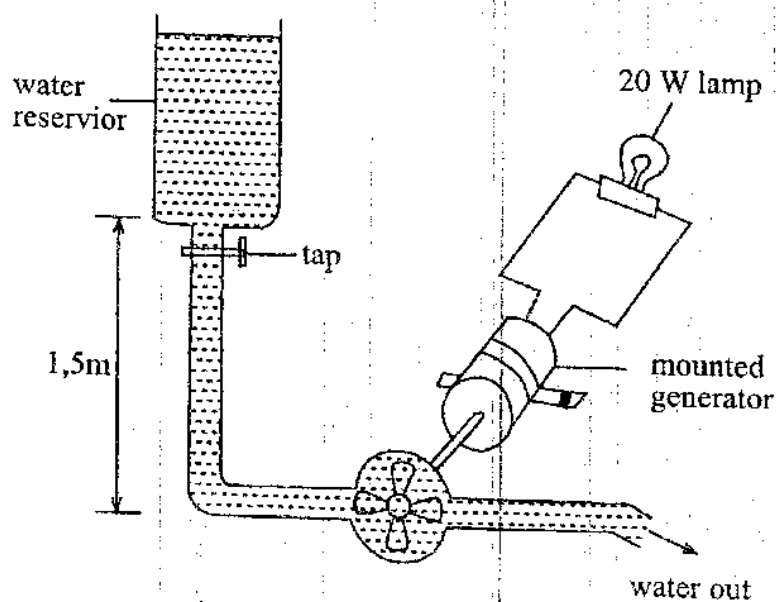


Fig.2.1

2.4 WORK, ENERGY & POWER P2&P3 QUESTIONS

Compiled by: R. Muga

- (i) Describe the main energy changes which take place when the tap is opened.
- (ii) The 20 W lamp operates normally when the tap is opened and water flows out at 3 kg s^{-1} .

Calculate the

1. potential energy lost by the water per unit time,
2. efficiency of the model.

[5]

- (c) Explain why a hydroelectric power station would be preferable to a thermal power station.

[2]

Q8:JUN 2016 P3

- 2 (a) (i) Define *work*.

- (ii) Show that

$$\text{power} = \text{force} \times \text{velocity}.$$

[3]

- (b) Fig. 2.1 shows a mineral ore dropping onto a long conveyor belt moving at a speed of 5 m s^{-1} . The mass of the conveyor belt is 50 kg.

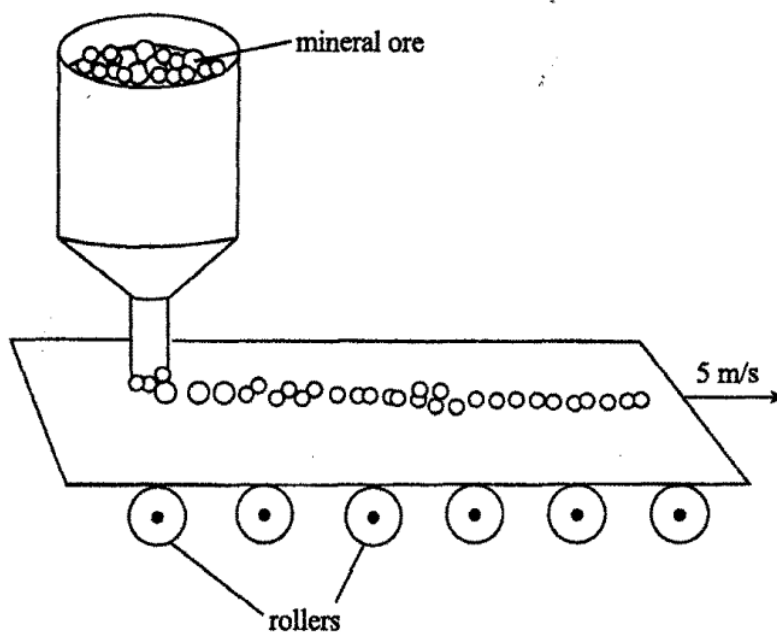


Fig. 2.1

The mineral ore is falling onto the belt at a rate of 750 kg s^{-1} .

- (i) Calculate the
1. net force required to maintain the steady speed of the belt,
 2. power used to move the belt.
- (ii) Explain why in practical situations, the power needed to maintain steady speed is greater than the value calculated in (b) (i) 2.

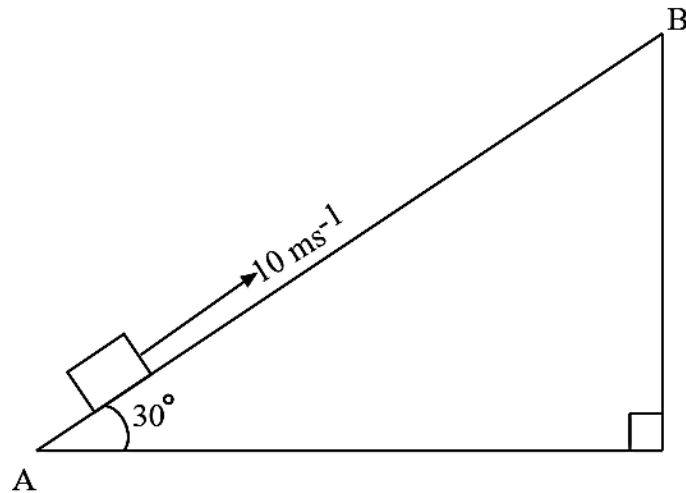
[7]

Q9:JUN 2017 P3

- 3 (a) Define *work*. [1]
- (b) (i) State **one** advantage of hydro power generation over thermal power generation.
- (ii) In a hydro-electric power station, water falls at a rate of $2.5 \times 10^5 \text{ kg s}^{-1}$ to drive turbines 220 m below.
- Calculate the loss in gravitational potential energy per second for the water as it reaches the turbines.
- (iii) The water leaves the turbines at a speed of 9.0 m/s.
- Calculate the maximum input power to the turbines, assuming that all the potential energy lost is transferred to the kinetic energy of the water. [7]
- (c) State and explain any **one** disadvantage of using hydro-electric power to produce electricity. [2]

Q10:2018 SPEC P3

- 1 (c) Fig. 1.3 shows a box of mass 3.0 kg projected up a rough incline at 30° to the horizontal with an initial velocity of 10 ms^{-1} . The velocity decreases to 4.0 ms^{-1} at B. The length of the plane is 4.0 m.

**Fig. 1.3**

- (i) Draw a free body diagram showing all forces acting on the box.
- (ii) Calculate the gain in gravitational potential energy.
- (iii) Determine the work done against friction.

[7]